



Examiners' Report Summer 2026

Pearson Edexcel GCE

In AS Further Mathematics (8FM0)

Paper 21 Further Pure Mathematics 1

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General Comments

Overall, this paper enabled learners across the full ability range to demonstrate their knowledge, understanding and problem-solving skills in the topics assessed.

The opening question was accessible to many learners and provided a familiar starting point. Question 2 appeared to be more challenging for some responses, particularly where learners were required to apply a second-order differential equation approximation.

The parabola questions provided opportunities for learners to demonstrate problem-solving skills. A range of valid methods was seen when finding the required coordinates, including approaches using parametric and Cartesian forms.

The vector question was well supported by the structure of the question and allowed learners to demonstrate their understanding of vector methods.

The final part of Question 5 was more demanding and gave stronger responses the opportunity to show clear reasoning.

Comments on Individual Papers

Question 1

This question was answered well, with most learners multiplying both sides of the inequality by $(3 - x)^2$ and then proceeded to collect all terms onto the left-hand side. In stronger responses, learners factorised out $(3 - x)$ formed a quadratic and then factorised this to obtain 3 critical values. In less successful responses, learners factorised the expression to form a cubic equation and then used a calculator to find the critical values. This method was acceptable, as learners were not relying entirely on their calculator and had shown an appropriate algebraic step before solving.

Another common approach was to rearrange to get both terms on one side and found a common denominator. This was followed multiplying by $(3 - x)$ and solving to find the critical values. Quite a few responses formed a fraction with a common denominator but didn't multiply through by $(3 - x)$. Instead, they solved the quadratic in the numerator by factoring and knew that the denominator meant that the value of 3 was not in the domain forming an asymptote and so obtained the three critical values.

After obtaining the three critical values, many learners sketched a cubic graph to identify the required region. In some responses, the cubic equation formed was treated as a positive cubic rather than a negative cubic, leading to an incorrect sketch. Learners are reminded that, when sketching graphs of polynomials, they should consider the sign of the leading coefficient, i.e., the coefficient of the highest power of x , as this determines the overall shape of the graph. Responses that used the correct cubic shape usually identified the required region, or the corresponding region appropriate to their critical values, including noting that $x = 3$ was not in the required region.

Question 2

This question appeared to be challenging for some learners. Less successful responses often gained the first mark for finding the second derivative when $t = 0$. Some learners did not apply the first and second order approximations correctly, using $V_{-1} = 200$, this meant that they did not gain any more marks.

Stronger responses formed and solved the simultaneous equations successfully. However, some responses stopped after finding the value of V_1 ,

rather than continuing to find the value of the house after one year. Learners needed to apply the approximation twice, using $h = 0.5$, to find the required value.

Many learners used $V_0 = 200,000$ instead of $V_0 = 200$. This was considered in the mark scheme and allowed these responses to access up to 6 of the 8 marks available. The final accuracy mark was a follow-through mark for $V_2 \times 1000$, rounded to the nearest thousand. Responses that had already worked in thousands did not gain this mark. Learners are reminded to check the definition of each variable carefully; in this question, V represented the value in thousands of pounds.

Stronger responses worked correctly with $V_0 = 200$, applied the approximation twice to find a value for V_2 , and then gave the value of the house in thousands of pounds. A common error was to use degrees rather than radians in the second-order differential equation. This did not affect the first mark, because $\sin 0 = 0$, but it did affect the use of $\sin 0.5$. These responses could still gain the follow-through accuracy mark. Learners are reminded that approximations involving differential equations should be carried out using radians.

Question 3

Part (a)

The aim of the question was for learners to apply the focus directrix property of parabolas, so the x coordinate $= 6a - a$ and then uses this value in the equation to find the y coordinate. Learners used a variety of methods to find the required coordinates using the parametric coordinates or cartesian coordinates with the equation of the curve and uses the length of $PS = 6a$. These methods showed the problem-solving abilities of the learners.

Part (b)

Learners who managed to find coordinates in part (a) went on successfully to find the value of a by using the area of the triangle. A common slip was simplifying $\frac{1}{2} \times 2a \times 2\sqrt{5}a$ to become $2\sqrt{5}a$ when simplified instead of $2\sqrt{5}a^2$. These responses were able to gain two of the three marks, with only the final accuracy mark not awarded.

Question 4

Most learners were able to score well with this question in parts (a) and (b).

Part (a)

Some responses correctly found the vectors $\overrightarrow{AB}$ and $\overrightarrow{AC}$ then attempted to find the cross product but omitted the fact that they needed to take the negative sign for the $\mathbf{j}$ component. Many learners scored all three marks.

Part (b)

Many responses found the vector $\overrightarrow{AD}$ and applied the correct formula for the volume of the tetrahedron, setting the volume equal to 30 to find a value for p . In some responses, the modulus in the volume formula was not fully considered, so the volume was used as being 30 only, leading to a negative value for p . Since the question stated that p was a positive constant, learners are reminded to check any conditions given in the question carefully. In this case, using the corresponding signed value of -30 for volume, led to the positive value for p .

Part (c)

Many responses gave a vector equation for the plane rather than the required Cartesian equation, and so did not gain marks for this part. Stronger responses found the Cartesian equation correctly and gained both marks.

Question 5

Part (a)

Most learners knew the t -formulae for $\sin x$ and $\cos x$ and substituted these into the left-hand side of the identity. Stronger responses showed the necessary algebraic stages to obtain the printed answer.

Part (b)

In less successful responses, learners were able to apply the result in part (a) to the equation and apply the t -formula for $\tan x$. For the next mark, they multiplied by $t(1 - t^2)$ and began to simplify. Some algebraic slips led to an incorrect quartic equation for t .

Some responses stopped at this point or used a calculator to find values of t . More successful responses continued by solving for t^2 . However, not all responses gave a reason why there were no real solutions.

Stronger responses gave a complete reason, explaining that since $t^2 < 0$, there are no real solutions. Learners are reminded that, where a conclusion is required, a clear mathematical reason should be given.